MAD 1 · PROJECT ID: 24F2007550

TrekManagement

Smart Trekking Management System

A full-stack web application that centralizes trekking expedition management — enabling administrators, guides, and trekkers to manage every stage of a trekking program through a single, secure digital platform.

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Course

Modern Application Development I (MAD 1)

Date

May 2026

Core Objectives

Digitize Management

Replace paper records with a centralized digital database

Online Booking

Real-time slot management and overbooking prevention

Guide Verification

Structured document upload and admin approval workflow

RBAC

Role-based access control for Admin, Guide, and Trekker

Booking History

Complete participant tracking and booking records

Companion Management

Manage group bookings and companion details seamlessly

Notifications

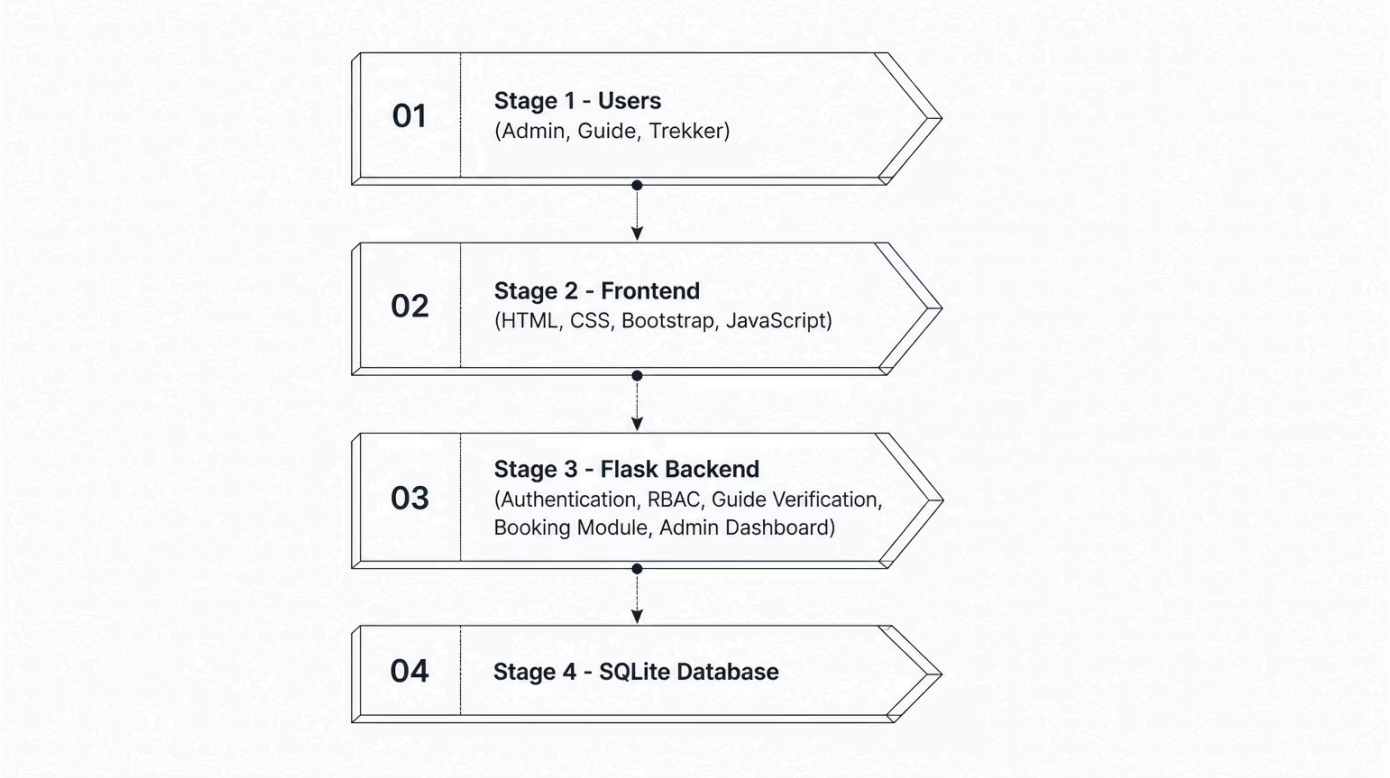
Real-time alerts for bookings, approvals, and updates

Admin Workflow

Streamlined approval and rejection processes

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System Architecture & Database Design



The layered architecture ensures clean separation of concerns — users interact with a responsive frontend, business logic is handled by Flask, and all data is persisted securely in SQLite.

Database Schema

Table	Purpose
Users	User authentication and role assignment
Treks	Trek details, schedules, and slot capacity
Bookings	Booking records and participant tracking
Documents	Guide verification file storage
Approval Requests	Guide approval and rejection workflow
Notifications	User alerts and system messages

Core Modules

Admin
- Dashboard & user management
- Guide approval workflow
- Trek management & notifications

Guide
- Verification form & document upload
- Track approval status
- Personal dashboard

Trekker
- Registration & trek browsing
- Book trek & companion details
- Booking history

Security Features

RBAC
Strict role-based access for Admin, Guide, and Trekker

Secure Login
Session management with validated credentials

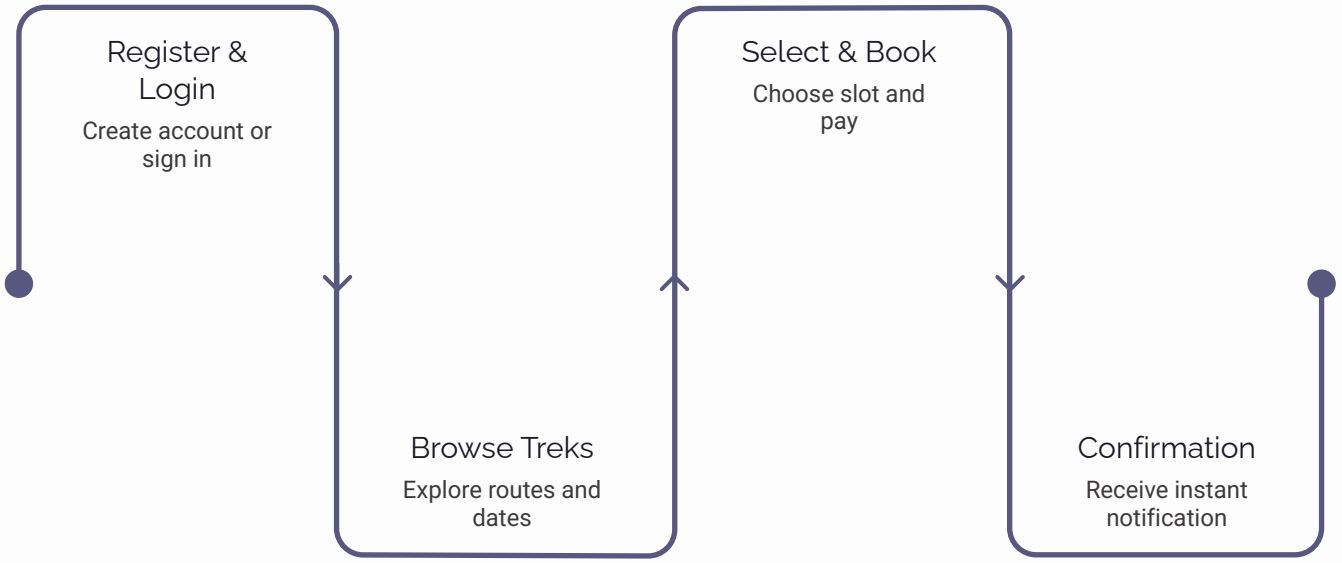
Guide Verification
Document-based identity and credential validation

Input Validation
Server-side sanitization to prevent injection attacks

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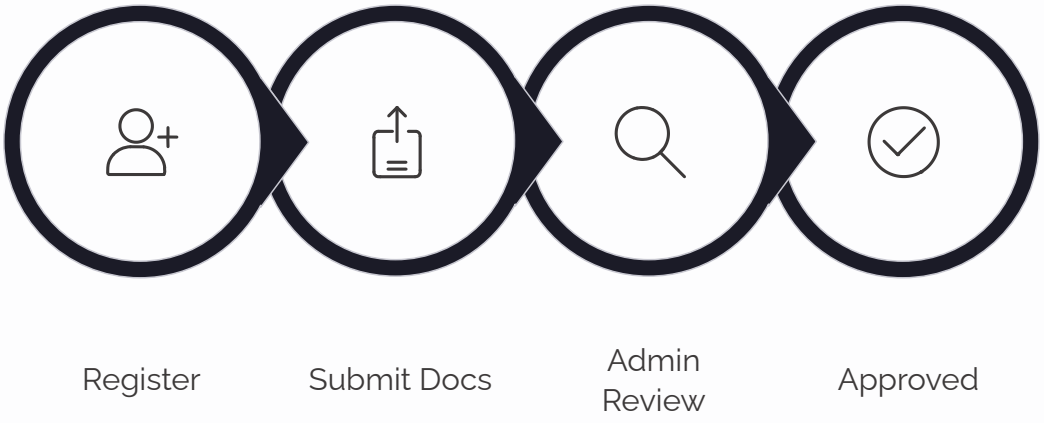
Workflows & Application Screens

Trek Booking Flow



Trekkers complete registration, browse available treks, select a slot, and receive instant confirmation — all within a seamless, guided flow.

Guide Approval Workflow



Guides submit credentials and documents for admin review. Upon approval, they gain access to the guide dashboard and can manage assigned treks.

REST API Endpoints

Method	Endpoint	Description
GET	`/api/treks`	Retrieve all available treks
POST	`/api/treks`	Add a new trek (Admin only)
PUT	`/api/treks/{id}`	Update trek details
DELETE	`/api/treks/{id}`	Delete a trek record
GET	`/api/bookings`	Retrieve booking list

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Highlights, Future Roadmap & Conclusion

Project Highlights

✔ RBAC
Role-based access control

✔ Guide Verification
Document-based workflow

✔ Online Booking
Real-time slot management

✔ Admin Dashboard
Centralized control panel

✔ Notifications
Real-time system alerts

✔ SQLite
Lightweight database

✔ REST APIs
Structured endpoints

✔ Responsive UI
Mobile-friendly design

Future Enhancements

AI Guide Recommendation
Intelligent matching of guides to treks based on skills and history

Weather API Integration
Real-time weather alerts for trek safety and planning

Google Maps Navigation
Route guidance and live location tracking for treks

Payment Gateway
Secure online payment processing for bookings

Mobile Application
Native iOS and Android app for on-the-go management

Emergency SOS
One-tap emergency alerts with location sharing

Predictive Risk Analysis
ML-based risk scoring for trek conditions and safety

QR Code Check-in
Fast, contactless participant check-in at trailheads

Conclusion

TrekManagement successfully demonstrates a secure, scalable, and user-friendly trekking management platform. By integrating authentication, guide verification, booking management, RBAC, and administrative workflows into a single application, the system provides an efficient solution for organizing trekking expeditions. The modular architecture supports future enhancements including AI-powered recommendations, predictive analytics, and mobile integration.

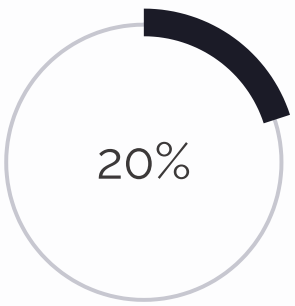
GitHub Repository

github.com/24f2007550/Trekking-Management-Application

Demo Video

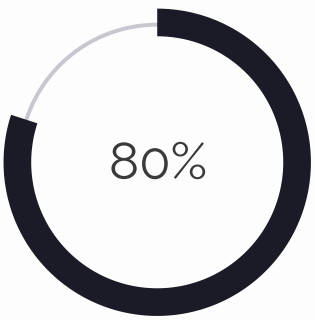
View on Google Drive

AI Usage Declaration



AI-Assisted

Documentation refinement, UI/UX suggestions, grammar, and formatting improvements



Author's Work

System architecture, business logic, RBAC design, database schema, backend implementation, and all core functionality

All major design decisions, application logic, workflows, and core functionality represent the original work of **Arshpreet Singh**. AI served only as a supporting tool for refinement and presentation quality.

Developed as the Trekking Management Application for MAD 1 (Modern Application Development I) — May 2026

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